# EduNova AI

### A Multimodal Student Learning, Doubt-Solving, and Hackathon Recommendation Platform

**Tagline:** Learn smarter. Solve anything. Discover opportunities.

**Capstone Title:**
**EduNova AI: A Multimodal AI-Powered Education Tracker, Multi-LLM Doubt Solver, and Personalized Hackathon Recommendation Platform**

## 1. Project Overview

EduNova AI is a Python-based web application designed for students. It combines three major features in one platform:

1. **Education Tracker** — Students can manage subjects, attendance, assignments, exams, study sessions, and academic progress.
2. **Multimodal AI Doubt Solver** — Students can upload PDFs, images, photos, screenshots, handwritten notes, or type a question. The platform sends the relevant content to multiple AI models and provides clear, grounded solutions.
3. **Live Hackathon Discovery and Recommendation** — The platform collects live hackathon listings from online sources and recommends suitable hackathons to each student based on skills, interests, level, location, and deadline. Students can directly open the official registration page.

The main goal of EduNova AI is to become a complete student companion: it helps students study, solve academic doubts, track progress, and find opportunities to grow.

## 2. Problem Statement

Students face several daily problems:

- They forget assignment deadlines, exam dates, and attendance requirements.
- They have study material in different formats: PDFs, photos, screenshots, handwritten notes, and lecture slides.
- They need explanations, summaries, quizzes, and solutions from their own study material.
- They use multiple AI tools separately, such as ChatGPT, DeepSeek, Gemini, and Claude.
- They miss hackathons because announcements are scattered across many websites.
- They do not know which hackathon matches their skills and interests.

EduNova AI solves these problems by bringing learning, doubt-solving, progress tracking, and opportunity discovery into one platform.

## 3. Objectives

- Build a student-friendly web application using Python.
- Allow students to upload and organize academic material.
- Support PDF, image, photo, screenshot, handwritten note, and text-prompt inputs.
- Use Retrieval-Augmented Generation (RAG) to answer questions from uploaded material.
- Integrate multiple AI providers, including OpenAI, DeepSeek, Gemini, and Anthropic.
- Allow students to compare answers from different AI models.
- Track attendance, assignments, exams, marks, and study time.
- Predict exam readiness and identify weak subjects.
- Collect live hackathon listings from online sources.
- Recommend hackathons using a personalized matching system.
- Provide a direct registration link for every recommended hackathon.
- Deploy the platform using Docker and a cloud service.

## 4. Key Features

### 4.1 Student Dashboard

The dashboard gives students a quick view of their academic status.

- Today’s study plan
- Upcoming assignments and exams
- Attendance percentage
- Subject-wise progress
- Study streak
- Pending doubts
- Recommended hackathons
- Deadline alerts

### 4.2 Education Tracker

Students can add and manage:

- Subjects
- Chapters and topics

- Assignments
- Exams
- Attendance
- Internal marks
- Study sessions
- Revision progress
- Notes and PDFs

The system can calculate:

- Subject completion percentage
- Attendance risk
- Assignment pending status
- Exam countdown
- Study consistency
- Predicted exam readiness

### 4.3 Multimodal AI Doubt Solver

Students can upload:

- PDF notes
- Textbooks
- Lecture slides
- Photos of handwritten notes
- Images of questions
- Screenshots of diagrams or code
- Text prompts

The app processes the input and provides:

- Direct answers
- Step-by-step explanations
- Summaries
- Important points
- Flashcards
- MCQs
- Short-answer questions
- Revision notes

### 4.4 Multi-LLM Answer Comparison

Students can choose one AI model or compare multiple models:

- ChatGPT / OpenAI
- DeepSeek
- Gemini
- Claude / Anthropic
- Optional local open-source model

All selected models receive the same student question and the same retrieved context from the uploaded material. This allows students to compare explanations and choose the best answer.

### 4.5 PDF and Image Intelligence

For PDFs:

- Extract text
- Split into chunks
- Create embeddings
- Store in a vector database
- Retrieve relevant sections for a question
- Generate answers with page references

For images and photos:

- Use OCR for handwritten or printed text
- Use vision-language models for diagrams, charts, screenshots, and question images
- Extract text and visual context
- Generate explanations or solutions

### 4.6 Live Hackathon Page

The Hackathons page displays live opportunities collected from online sources.

Each hackathon card includes:

- Hackathon title
- Organizer
- Description
- Mode: online, offline, or hybrid
- Start date and registration deadline
- Prize or rewards
- Themes and tags

- Eligibility
- Source platform
- Official registration link

Students can:

- Search hackathons
- Filter by AI, web, mobile, open source, India, online, prize, and deadline
- Save hackathons
- View hackathon details
- Click **Register Now** to open the official registration page

### 4.7 Personalized Hackathon Recommendation

EduNova AI recommends hackathons using a student profile.

It considers:

- Skills
- Interests
- Experience level
- Preferred mode
- Location preference
- Deadline urgency
- Past saved or registered hackathons

For every recommendation, the app shows a match score and explains why the hackathon is suitable.

Example:

> **Recommended: AI for Education Hackathon**
> **Match Score:** 88%
> **Why recommended:**
>
> - Matches your Python, Machine Learning, and Streamlit skills.
> - Focuses on education, one of your selected interests.
> - Online format matches your preference.
> - Registration closes in 5 days.
> - Suitable for intermediate-level students.

### 4.8 Quiz and Revision Generator

From uploaded PDFs or notes, the app can generate:

- Multiple-choice questions
- Short-answer questions
- Flashcards
- Chapter summaries
- Last-minute revision sheets
- Important formulas and definitions

### 4.9 Notifications and Reminders

The system can remind students about:

- Assignment deadlines
- Exam dates
- Low attendance
- Pending revision topics
- Hackathon registration deadlines
- Hackathon submission deadlines

## 5. System Architecture

```
Student Browser
      |
Streamlit / React Frontend
      |
FastAPI Backend
      |
-----------------------------------------------
|                Core Services                |
-----------------------------------------------
| Authentication | Education Tracker | AI Router |
-----------------------------------------------
      |                   |                |
PDF/Image Processor   Vector Database   Hackathon Collector
      |                   |                |
Text Extraction       Embeddings       External Sources
OCR / Vision Model    RAG Retrieval    Devpost / Unstop etc.
      |                   |                |
-----------------------------------------------
|              Database: PostgreSQL / SQLite    |
-----------------------------------------------
```

## 6. Technology Stack

| Layer | Technology |
|---|---|
| Programming Language | Python |
| Frontend | Streamlit for capstone demo; React optional for production |
| Backend | FastAPI |
| Database | SQLite for capstone; PostgreSQL for production |
| PDF Processing | pypdf, pdfplumber, PyMuPDF |
| OCR | Pytesseract, PaddleOCR |
| Image Understanding | Vision-language model APIs |
| Embeddings | OpenAI embeddings, Gemini embeddings, or open-source embedding models |
| Vector Database | FAISS, Chroma, or Qdrant |
| AI Models | OpenAI, DeepSeek, Gemini, Anthropic |
| Recommendation | TF-IDF, cosine similarity, rule-based scoring |
| Authentication | JWT or session-based authentication |
| Task Scheduling | APScheduler or Celery |
| Deployment | Docker, Render, Hugging Face Spaces, or college server |
| Version Control | Git and GitHub |

## 7. AI Workflow

### 7.1 PDF Question Answering

1. Student uploads a PDF.
2. Backend extracts text using a PDF parser.
3. Text is cleaned and split into smaller chunks.
4. Each chunk is converted into an embedding.
5. Embeddings are stored in a vector database.
6. When the student asks a question, the system retrieves the most relevant chunks.
7. The retrieved chunks and question are sent to the selected AI model.
8. The model generates an answer using only the provided context.
9. The answer is displayed with page references where possible.

### 7.2 Image or Photo Question Answering

1. Student uploads an image or photo.
2. If the image contains text, OCR extracts it.
3. If the image contains a diagram, chart, handwritten question, or screenshot, a vision-language model analyzes it.
4. The extracted text and visual description are combined.
5. The content is sent to the selected AI model.
6. The model returns an explanation or solution.

### 7.3 Multi-Model Comparison

1. Student asks one question.
2. Student selects two or more AI providers.
3. Backend creates the same RAG prompt for each provider.
4. Each provider returns an answer.
5. Frontend displays answers side by side.
6. Student can save the best answer to their notes.

## 8. Hackathon Recommendation Logic

### 8.1 Student Profile

The system stores:

- Skills
- Interests
- Experience level
- Preferred mode
- Location
- Availability
- Saved hackathons
- Registered hackathons

### 8.2 Hackathon Profile

For every hackathon, the system stores:

- Title
- Description
- Themes

- Required skills
- Mode
- Location
- Deadline
- Prize
- Eligibility
- Registration URL
- Source platform

### 8.3 Match Score Formula

```
Match Score =
(0.35 × Skill Match) +
(0.25 × Interest Match) +
(0.15 × Deadline Urgency) +
(0.10 × Mode/Location Match) +
(0.10 × Experience-Level Match) +
(0.05 × Past Behavior Match)
```

### 8.4 Recommendation Explanation

After calculating the score, the system can use an LLM to generate a short explanation:

```
Explain in 3 simple bullet points why this hackathon is recommended
for this student. Use only the provided student profile and hackathon
details. Do not invent facts.
```

## 9. Database Design

### users

- user_id
- name
- email
- password_hash
- skills
- interests
- preferred_mode
- location
- created_at

### subjects

- subject_id
- user_id
- subject_name
- total_topics
- completed_topics
- attendance_percentage
- created_at

### assignments

- assignment_id
- user_id
- subject_id
- title
- description
- due_date
- status
- created_at

### exams

- exam_id
- user_id
- subject_id
- exam_name
- exam_date
- syllabus
- status

### study_sessions

- session_id
- user_id
- subject_id
- topic
- duration_minutes
- study_date

- notes

### documents

- document_id
- user_id
- subject_id
- file_name
- file_type
- file_url
- uploaded_at

### document_chunks

- chunk_id
- document_id
- chunk_text
- page_number
- embedding_id

### chat_messages

- message_id
- user_id
- document_id
- question
- answer
- ai_provider
- created_at

### hackathons

- hackathon_id
- title
- description
- organizer
- source
- source_url
- registration_url
- mode

- location
- start_date
- deadline
- prize
- tags
- status
- last_updated

### saved_hackathons

- saved_id
- user_id
- hackathon_id
- saved_at

## 10. API Endpoints

### Authentication

- `POST /api/auth/register`
- `POST /api/auth/login`
- `GET /api/auth/me`

### Education Tracker

- `POST /api/subjects`
- `GET /api/subjects`
- `POST /api/assignments`
- `GET /api/assignments`
- `POST /api/exams`
- `GET /api/exams`
- `POST /api/study-sessions`
- `GET /api/dashboard`

### Documents and AI

- `POST /api/documents/upload`
- `GET /api/documents`
- `POST /api/ai/ask`

- `POST /api/ai/compare`
- `POST /api/ai/quiz`
- `POST /api/ai/summary`

### Hackathons

- `GET /api/hackathons`
- `GET /api/hackathons/recommendations`
- `POST /api/hackathons/save`
- `GET /api/hackathons/saved`

## 11. User Flow

1. Student registers and logs in.
2. Student adds subjects, exams, assignments, and skills.
3. Student uploads PDFs, images, or notes.
4. Student asks a doubt or requests a summary/quiz.
5. EduNova AI retrieves relevant content and sends it to selected AI models.
6. Student receives answers, explanations, and references.
7. Student visits the Hackathons page.
8. EduNova AI shows live hackathons and personalized recommendations.
9. Student clicks **Register Now** to open the official registration page.
10. Student tracks progress and receives reminders.

## 12. Security and Privacy

- Store passwords using secure hashing.
- Keep AI API keys in environment variables, never in source code.
- Use HTTPS in production.
- Validate uploaded file types and size.
- Store user documents securely.
- Allow users to delete uploaded documents and chat history.
- Use official APIs where possible.
- Respect website terms, rate limits, and robots.txt when collecting hackathon data.
- Cache hackathon data instead of calling external websites repeatedly.

## 13. Evaluation Metrics

| Feature | Metric |
|---|---|
| PDF Question Answering | Answer relevance, retrieval precision, citation accuracy |
| OCR | Character error rate, text extraction accuracy |
| Multi-LLM Comparison | Response quality, latency, user preference |
| Exam Readiness Prediction | Accuracy, F1-score, ROC-AUC |
| Hackathon Recommendation | Precision@K, recall@K, click-through rate |
| Education Tracker | Task completion rate, reminder effectiveness |
| System Performance | API response time, uptime, error rate |
| User Experience | Student satisfaction survey |

## 14. Future Enhancements

- Mobile application using React Native or Flutter.
- Voice-based doubt asking.
- Live lecture recording and automatic summarization.
- AI-generated personalized revision timetable.
- Teammate finder for hackathons.
- GitHub profile analysis for better hackathon recommendations.
- Local offline AI model using Ollama.
- Collaboration spaces for group study.
- Badge and reward system for study consistency.
- Integration with Google Calendar and WhatsApp reminders.

## 15. Expected Outcome

EduNova AI will help students:

- Organize their academic life in one place.
- Get answers from their own PDFs, images, and notes.
- Compare responses from multiple AI models.
- Identify weak subjects and improve exam preparation.
- Discover relevant live hackathons.
- Receive clear reasons for hackathon recommendations.
- Register for opportunities directly through official links.

## 16. Conclusion

EduNova AI is a practical, advanced, and student-focused capstone project. It combines education tracking, multimodal AI, retrieval-augmented generation, multi-LLM comparison, and personalized hackathon recommendation. The project is useful in daily student life and demonstrates strong Python, AI/ML, backend, database, and deployment skills.